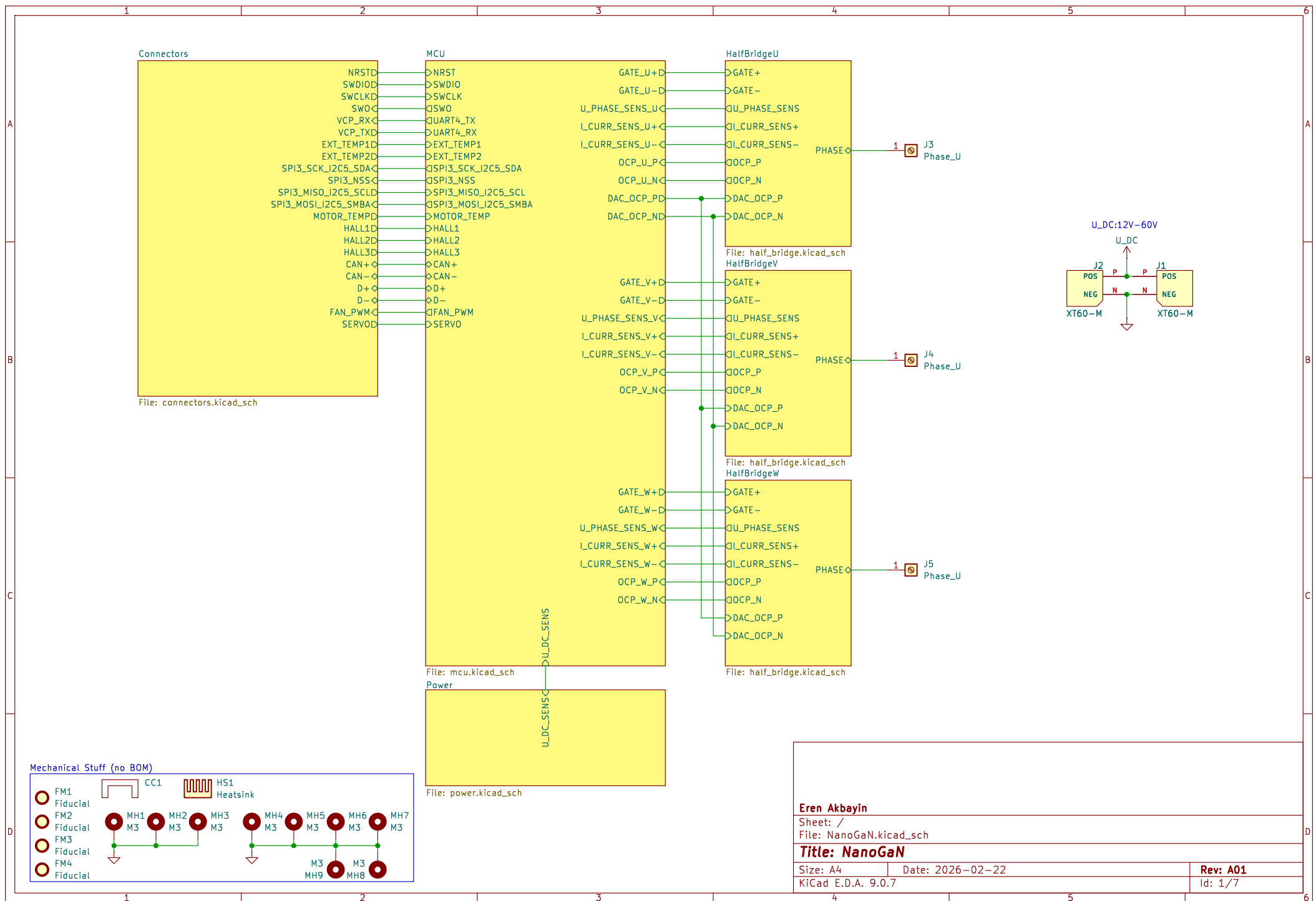
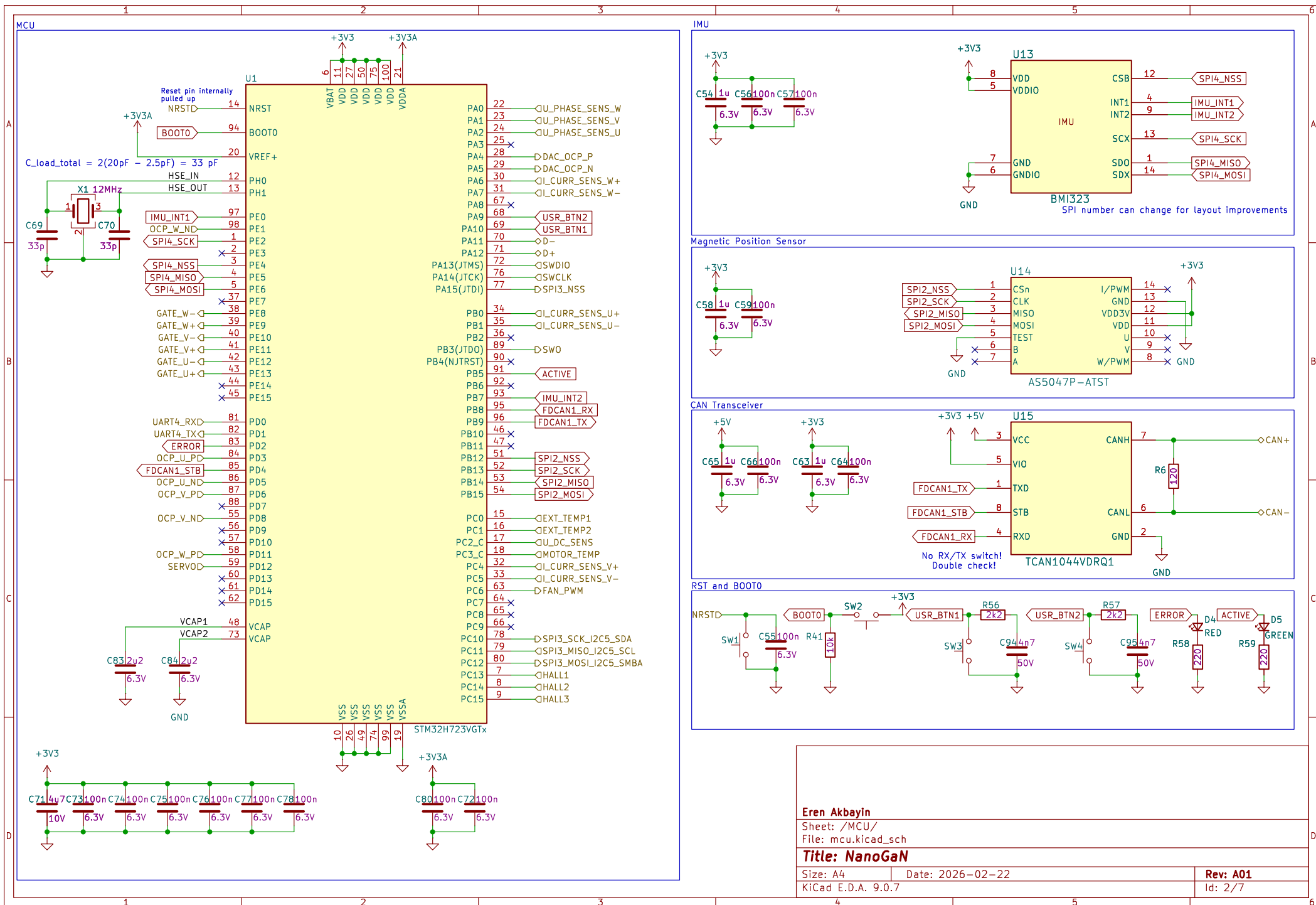
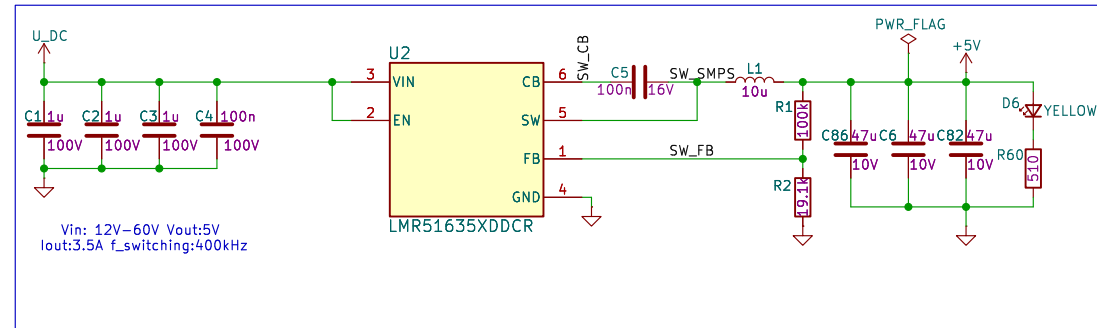


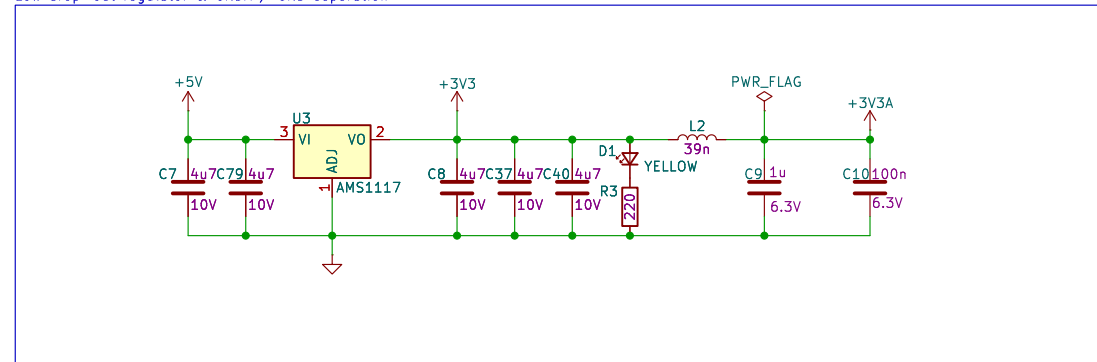




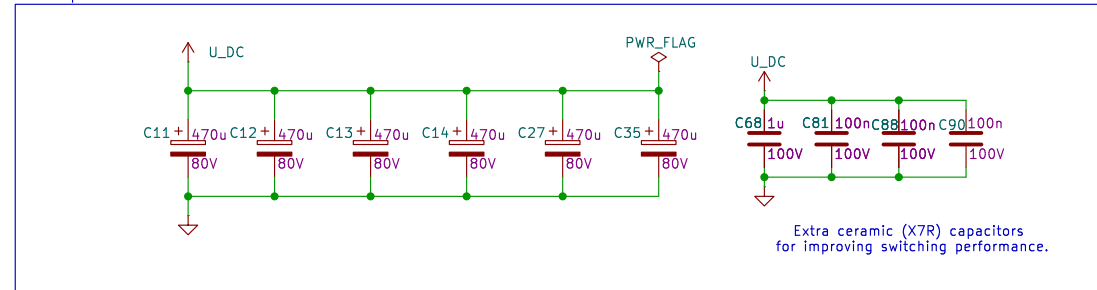
Buck Converter



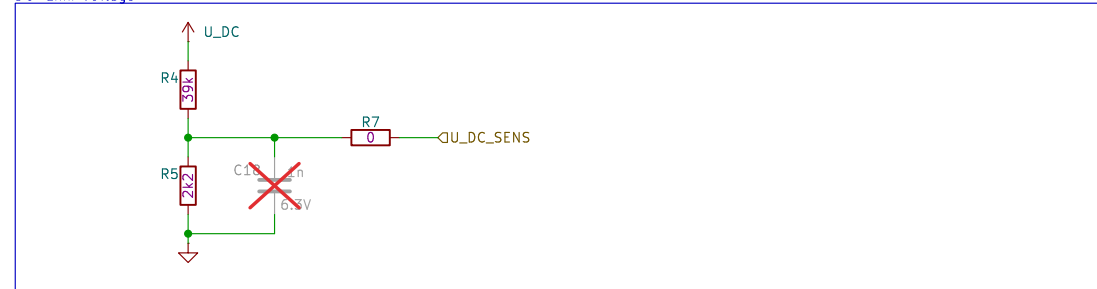
Low drop-out regulator & GNDA / GND separation



Bulk Capacitors



DC-Link Voltage

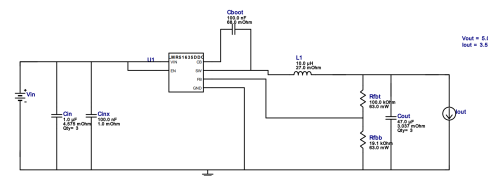


WEBENCH[®] Design Report

Design : 29 LMR51635DDC
LMR51635DDC 12V-60V to 5.00V @ 3.5A

VinMin = 12.0V
VinMax = 60.0V
Vout = 5.0V
Iout = 3.5A

Device = LMR51635DDC
Topology = Buck
Created = 2026-01-18 17:57:50.631
BOM Cost = \$2.43
BOM Count = 12
Total Pd = 2.47W



Eren Akbayin

Sheet: /Power/
File: power.kicad_sch

Title: NanoGaN

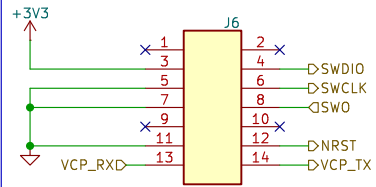
Size: A4	Date: 2026-02-22
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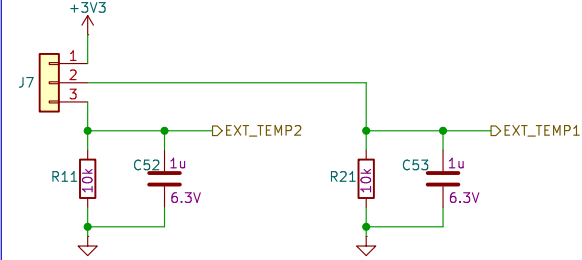
Rev: A01

Id: 3/7

ST-Link V3 MINIE CONNECTOR (STDC14)

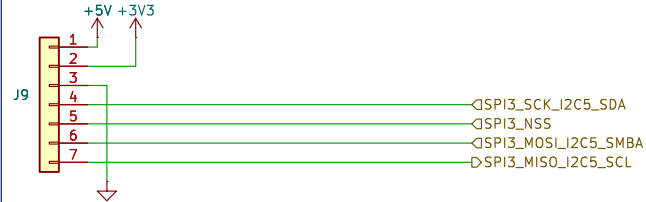


External Temp. Readings

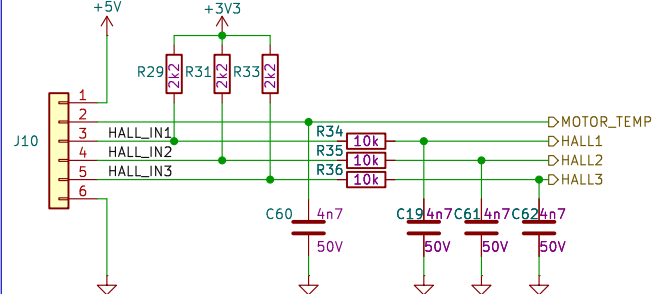


for motor and etc.

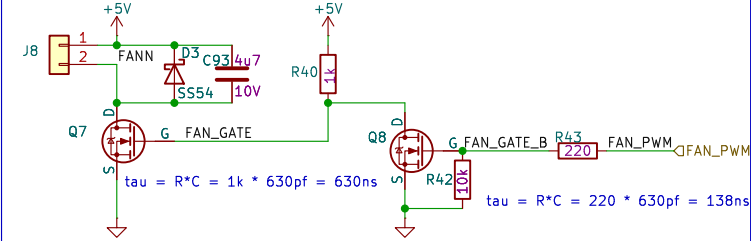
SPI/I2C/USART connection



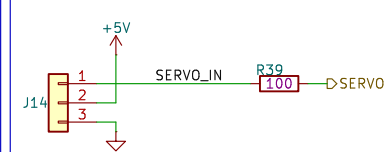
Hall connector and filtering



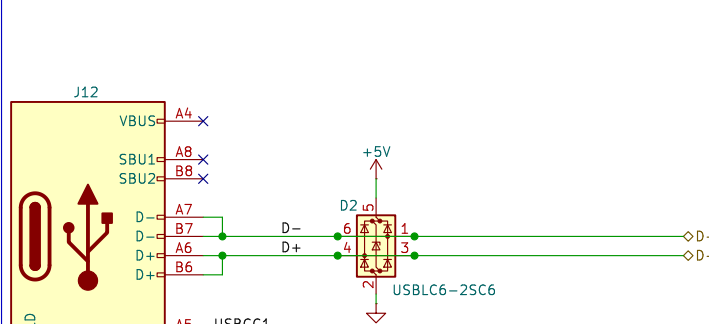
Fan connector



Servo



USB-C Connector



Differential Pair settings

Layers: 4

PCB Thickness: 0.8

Inner Copper Weight: 0.3oz

Outer Copper Weight: 1oz

Unit: mm

Impedance Configure

+ New Impedance

+ Duplicate Impedance

+ Calculate

Impedance (Ω)	Type	Signal Layer	Top Ref	Bottom Ref	Trace Spacing (mm)	Impedance trace to copper (mm)
90	Differential Pair (Non-coupled)	L1	/	L2	0.20	/
100	Differential Pair (Non-coupled)	L8	L2	/	0.5	/

JLC0808H-2310 (Standard) (Finished thickness: 0.74±0.10mm)

JLC0808H-1080 (特例) (成品厚度: 0.59±0.10mm)

Impedance (Ω)	Type	Signal Layer	Top Ref	Bottom Ref	Trace Width	Trace Spacing	Impedance trace to copper
L1	Outer Copper Weight	1.38			0.0300		
Prepreg	0.075mm VMCZ without copper	3.91			0.0994		
L2	Inner Copper Weight	0.60			0.0750		
Core	0.075mm VMCZ without copper	2.95			0.0750		
L3	Inner Copper Weight	0.60			0.0994		
Prepreg	0.075mm VMCZ without copper	3.91			0.0994		
L4	Inner Copper Weight	0.60			0.0750		
Core	0.075mm VMCZ without copper	2.95			0.0750		
L5	Inner Copper Weight	0.60			0.0994		
Prepreg	0.075mm VMCZ without copper	3.91			0.0994		
L6	Outer Copper Weight	1.38			0.0300		

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Sheet: /Connectors/
File: connectors.kicad_sch

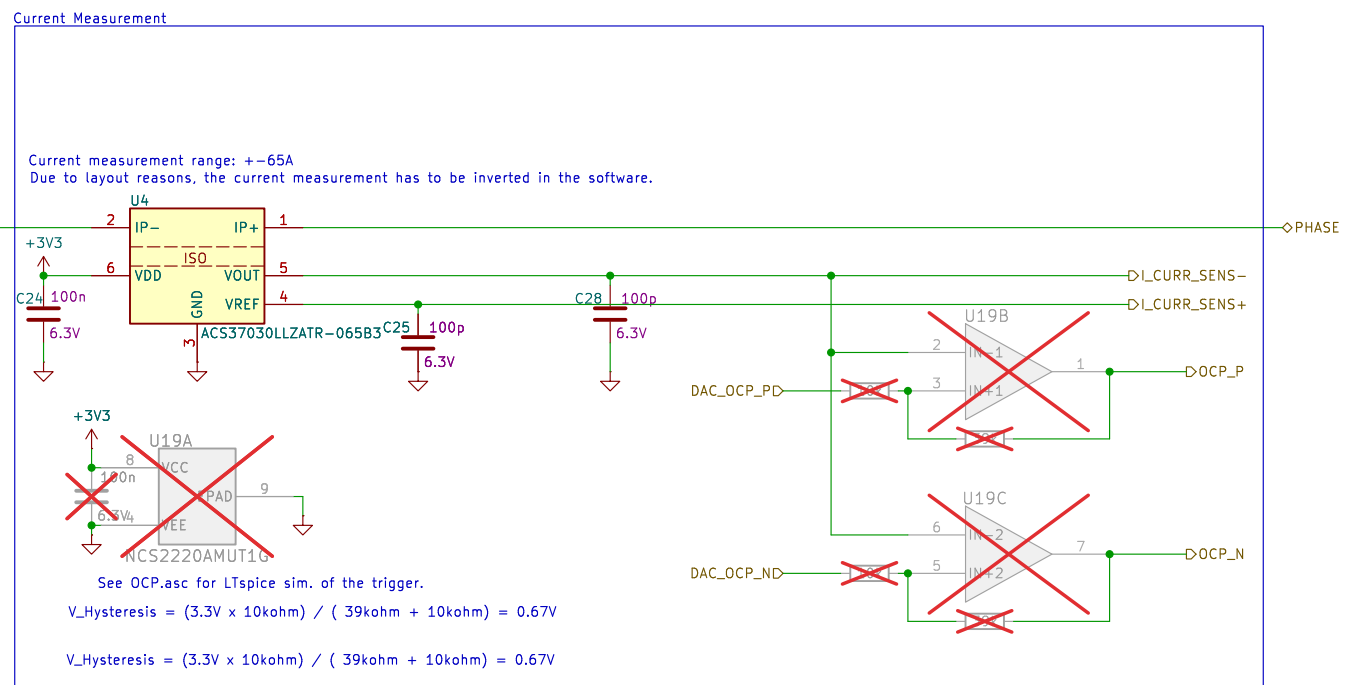
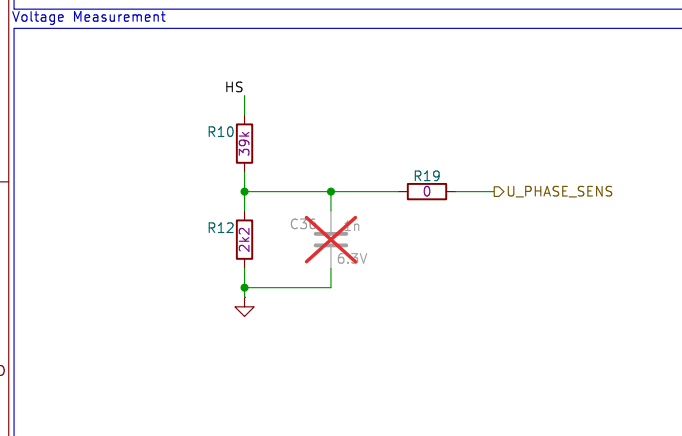
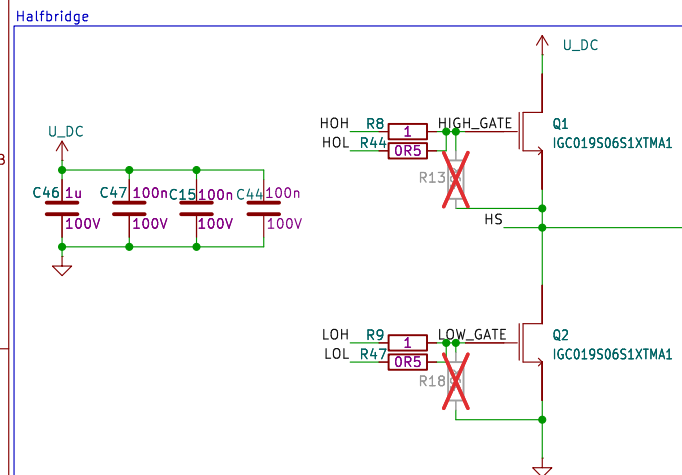
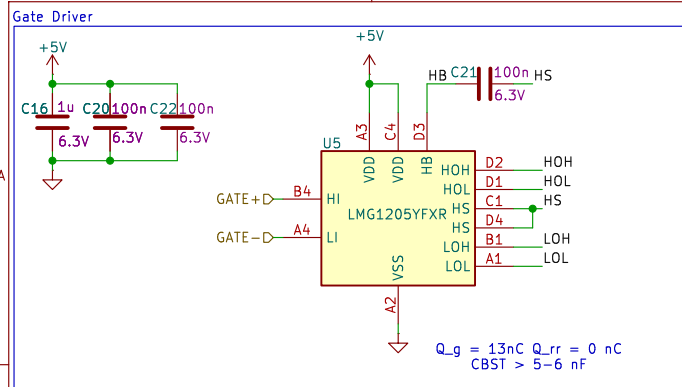
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Size: A4 Date: 2026-02-22

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Rev: A01

Id: 4/7



Eren Akbayin

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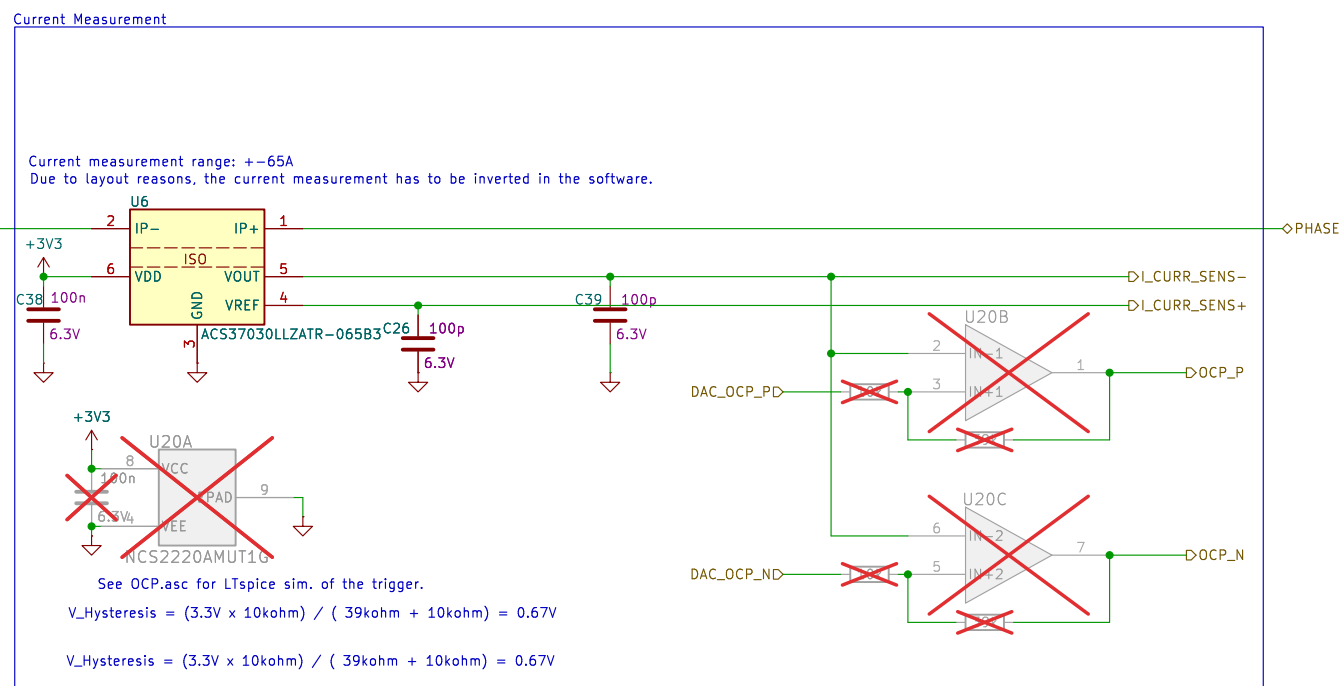
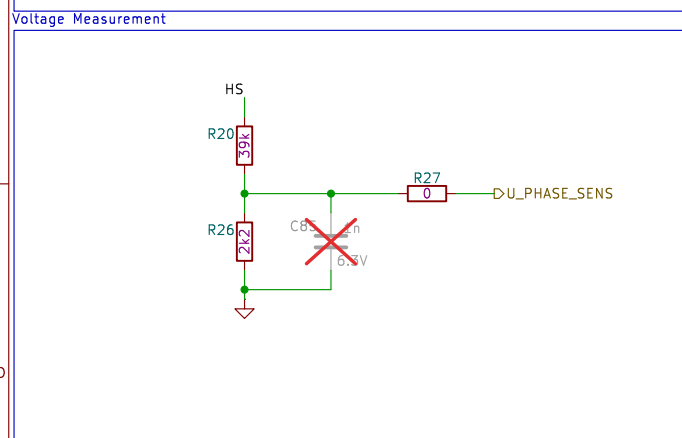
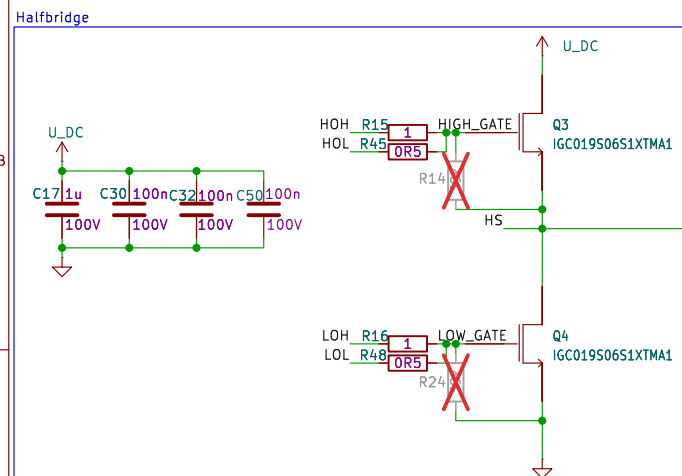
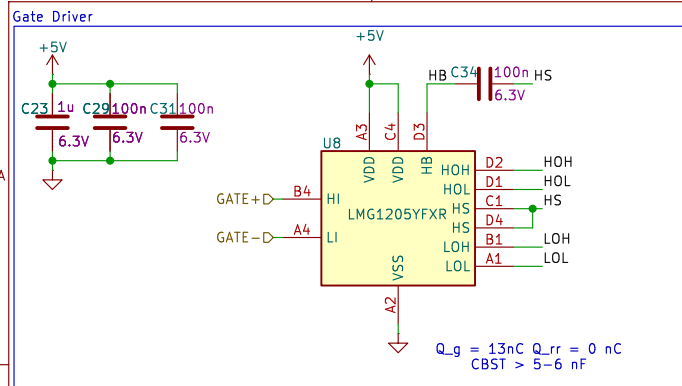
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Size: A4	Date: 2026-02-22
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Rev: A01

Id: 5/7



Eren Akbayin

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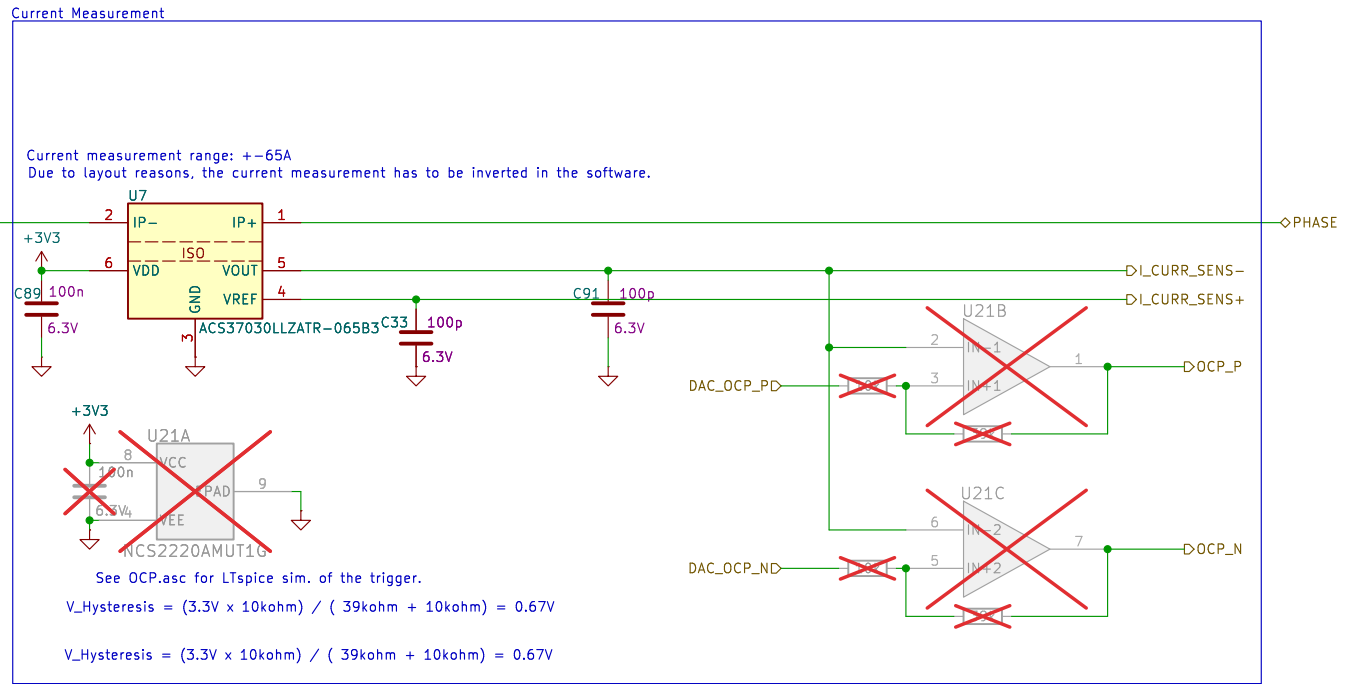
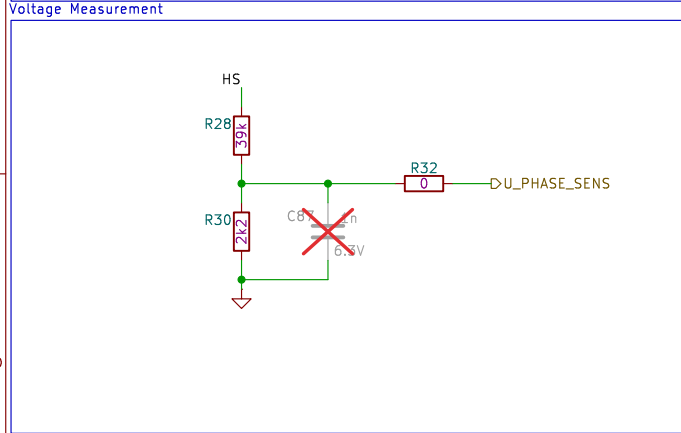
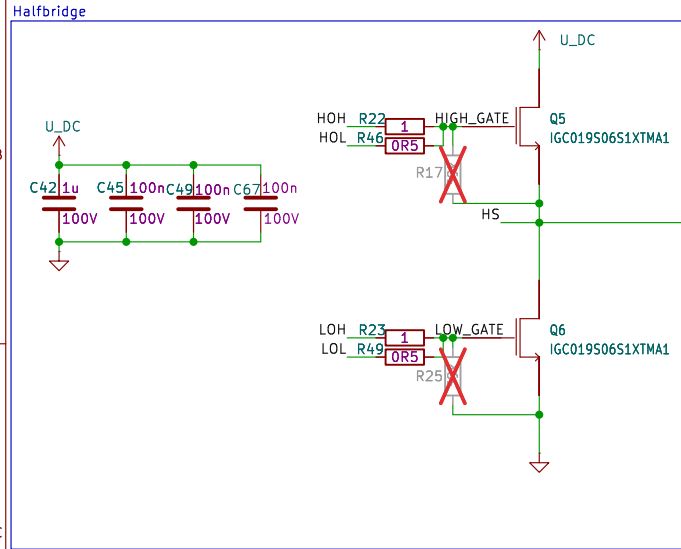
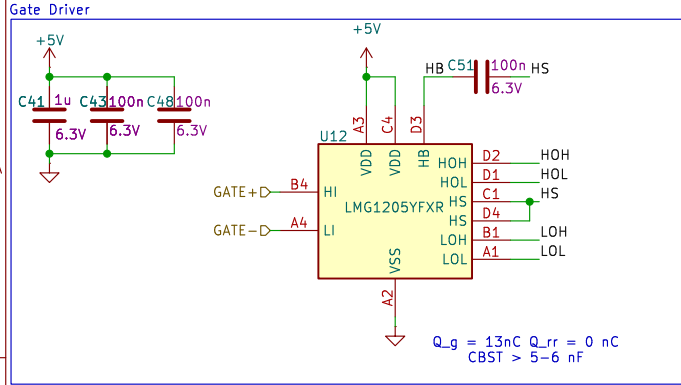
Title: NanoGaN

Size: A4	Date: 2026-02-22
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Rev: A01

Id: 6/7



Eren Akbayin

Sheet: /HalfBridgeW/
File: half_bridge.kicad_sch

Title: NanoGaN

Size: A4 Date: 2026-02-22
KiCad E.D.A. 9.0.7

Rev: A01
Id: 7/7